

10.5.1: Scattering amplitude (Relation between scattering amplitude and differential cross-section): Consider the elastic scattering of two spinless and non-relativistic particles of masses m_1 and m_2 . If the interaction between particles depends only on their relative distance r , then Schrodinger wave equation is given by,

$$\nabla^2 \psi(\vec{r}) + \frac{2\mu}{\hbar^2} \{E - V(\vec{r})\} \psi(\vec{r}) = 0$$

The reduced mass μ given by,

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} \Rightarrow \frac{1}{\mu} = \frac{m_1 + m_2}{m_1 m_2} \Rightarrow \mu = \frac{m_1 m_2}{m_1 + m_2}$$

Complete solution of this equation consists of two parts:

(i)-Incident wave function and

(ii)-Scattered wave function

For a finite range a of potential $V(r)$, the interaction between the particle and the potential occurs only in a limited region of space $r \leq a$, called the range of potential.

Outside the range $r > a$, $V(r) = 0$ and above equation then becomes;

$$\nabla^2 \phi_{inc}(\vec{r}) + \frac{2\mu E}{\hbar^2} \phi_{inc}(\vec{r}) = 0 \Rightarrow \left(\nabla^2 + \frac{2\mu E}{\hbar^2} \right) \phi_{inc}(\vec{r}) = 0$$

Put $2\mu E/\hbar^2 = k_0^2$ to get,

$$(\nabla^2 + k_0^2) \phi_{inc}(\vec{r}) = 0$$

Solution of this equation is given by a plane wave,

$$\phi_{inc}(\vec{r}) = e^{i\vec{k}_0 \cdot \vec{r}}$$

The scattered wave is,

$$\phi_{sc}(\vec{r}) = f(\theta, \varphi) \frac{e^{i\vec{k} \cdot \vec{r}}}{r}$$

Here $f(\theta, \varphi)$ is known the scattering amplitude

$\vec{k}$ is the wave vector of the scattered particle, and

θ is the angle between $\vec{k}_0$ and $\vec{k}$

Incident flux density is given by,

$$J_{inc} = \frac{i\hbar}{2\mu} (\phi_{inc} \nabla \phi_{inc}^* - \phi_{inc}^* \nabla \phi_{inc}) = \frac{i\hbar}{2\mu} (e^{i\vec{k}_0 \cdot \vec{r}} e^{-i\vec{k}_0 \cdot \vec{r}} \times -ik_0 - e^{-i\vec{k}_0 \cdot \vec{r}} e^{i\vec{k}_0 \cdot \vec{r}} ik_0) \\ \Rightarrow J_{inc} = \frac{i\hbar}{2\mu} (-ik_0 - ik_0) = \frac{i\hbar}{2\mu} (-2ik_0) = \frac{\hbar k_0}{\mu}$$

Scattered flux density is given by,

$$J_{sc} = \frac{i\hbar}{2\mu} (\phi_{sc} \nabla \phi_{sc}^* - \phi_{sc}^* \nabla \phi_{sc}) \\ \Rightarrow J_{sc} = \frac{i\hbar}{2\mu} \left(f(\theta, \varphi) \frac{e^{i\vec{k} \cdot \vec{r}}}{r} f^*(\theta, \varphi) \frac{e^{-i\vec{k} \cdot \vec{r}}}{r} \times -ik - f^*(\theta, \varphi) \frac{e^{-i\vec{k} \cdot \vec{r}}}{r} f(\theta, \varphi) \frac{e^{i\vec{k} \cdot \vec{r}}}{r} ik \right) \\ \Rightarrow J_{sc} = \frac{i\hbar}{2\mu r^2} (-ik|f(\theta, \varphi)|^2 - ik|f(\theta, \varphi)|^2) = \frac{i\hbar}{2\mu r^2} (-2ik|f(\theta, \varphi)|^2) = \frac{\hbar k}{\mu r^2} |f(\theta, \varphi)|^2$$

The differential cross-section is given by relation,

$$\frac{d\sigma}{d\Omega} = \frac{1}{J_{inc}} \frac{dN}{d\Omega}$$

The number $dN(\theta, \varphi)$ of particles scattered into an element of solid angle $d\Omega$ in the direction (θ, φ) passing through a surface element dA per unit time is;

$$\begin{aligned} dN &= J_{sc} dA = J_{sc} r^2 d\Omega = \frac{\hbar k}{\mu r^2} |f(\theta, \varphi)|^2 r^2 d\Omega = \frac{\hbar k}{\mu} |f(\theta, \varphi)|^2 d\Omega \\ \Rightarrow \frac{dN}{d\Omega} &= \frac{\hbar k}{\mu} |f(\theta, \varphi)|^2 \end{aligned}$$

Therefore, differential cross-section is,

$$\frac{d\sigma}{d\Omega} = \frac{1}{J_{inc}} \frac{\hbar k}{\mu} |f(\theta, \varphi)|^2 = \frac{1}{\hbar k_0 / \mu} \frac{\hbar k}{\mu} |f(\theta, \varphi)|^2 = \frac{k}{k_0} |f(\theta, \varphi)|^2$$

For elastic scattering $k = k_0$, so

$$\frac{d\sigma}{d\Omega} = |f(\theta, \varphi)|^2$$

Thus differential cross-section can be obtained by calculating scattering amplitude. Due to this, scattering amplitude has central role in scattering theory.